Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Wyeth Dairy, Inc.

Physical address of dairy:

7319 Beckwith RD	Modesto	Stanislaus	95358
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1960

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X012-X018-X006-XXXX X012-X018-X015-XXXX

B. OPERATORS

Wyeth, Ivan Jr

Operator name: Wyeth, Ivan Jr

Telephone no.: (209) 545-0891 (Landline) (Cellular)

7319 Beckwith RD	Modesto	CA	95358
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Wyeth, Ivan Jr

Legal owner name: Wyeth, Ivan Jr

Telephone no.: (209) 545-0891 (Landline) (Cellular)

7319 Beckwith RD	Modesto	CA	95358
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	584	60	245	245	80	0
Maximum number	584	60	245	245	80	0
Average number	584	60	245	245	80	0
Avg live weight (lbs)	1,400	1,450	1,000	700		

Predominant milk cow breed: Holstein

Average milk production: 80 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 21,704.88 *tons per reporting period*

Total nitrogen from manure: 266,362.22 *lbs per reporting period*

After ammonia losses (30% loss applied): 186,453.55 *lbs per reporting period*

Total phosphorus from manure: 44,409.86 *lbs per reporting period*

Total potassium from manure: 115,678.27 *lbs per reporting period*

Total salt from manure: 288,773.40 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 18,015,000 *gallons*

Total nitrogen generated: 62,612.16 *lbs*

Total phosphorus generated: 11,565.39 *lbs*

Total potassium generated: 71,767.46 *lbs*

Total salt generated: 576,495.97 *lbs*

	18,015,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	18,015,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
MID Lateral 6	Surface water
MID Lateral 7	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
07/19/2023	Solid commercial fertilizer *UN32*	19.60 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	12,531.46	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	12,531.46	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Bacon	32	32	2	manure	X012-X019-X002-XXXX
Boer	9	9	2	manure	X003-X011-X001-XXXX
Connacher	44	44	2	process wastewater	X012-X022-X009-XXXX X012-X022-X016-XXXX
Cover	19	19	2	process wastewater	X012-X018-X003-XXXX
Joliff 1	34	34	2	process wastewater	X012-X018-X015-XXXX
Joliff 2	12	12	2	process wastewater	X012-X018-X015-XXXX
Mattos	18	18	2	manure	X012-X034-X004-XXXX
N. Jackson	19	19	2	process wastewater	X012-X018-X006-XXXX
S. Jackson	9	9	2	process wastewater	X012-X018-X006-XXXX
Totals for areas that were used for application	196	196	18		
Totals for areas that were not used for application					
Land application area totals	196	196	18		

B. CROPS AND HARVESTS

Bacon

Field name: Bacon

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 32 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/19/2023	519.93 *ton*	As-is		64.2	4,100.00	1,200.00	9,800.00		8.58

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	16.25	133.23	38.99	318.46	998.15

Bacon

06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 32 Plant date: 06/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/03/2023	928.38 *ton*	As-is		65.2	3,100.00	1,100.00	3,900.00		5.52

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.01	179.87	63.83	226.29	1,114.61

Boer

Field name: Boer

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 9 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	103.16 *ton*	As-is		72.6	5,600.00	1,300.00	10,500.00		11.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	11.46	128.38	29.80	240.71	728.63

06/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 9 Plant date: 06/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/03/2023	232.87 *ton*	As-is		67.2	4,000.00	1,000.00	3,400.00		4.67

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.87	207.00	51.75	175.95	792.67

Connacher

Field name: Connacher

11/09/2022: Oats, hay

Crop: Oats, hay Acres planted: 44 Plant date: 11/09/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/26/2023	183.00 *ton*	As-is		16.6	24,000.00	2,160.00	19,900.00		8.86

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	160.00	26.00	132.00	0.00
Total actual harvest content	4.16	199.64	17.97	165.53	614.65

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 44 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/03/2023	1,431.36 *ton*	As-is		65.2	4,600.00	1,400.00	6,700.00		7.02

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.53	299.28	91.09	435.91	1,589.43

Cover

Field name: Cover

11/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 19 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/18/2023	359.97 *ton*	As-is		72.6	5,600.00	1,300.00	10,500.00		11.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	18.95	212.19	49.26	397.86	1,204.35

Cover

06/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 19 Plant date: 06/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/08/2023	599.45 *ton*	As-is		66.2	4,200.00	1,300.00	5,200.00		6.53

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	31.55	265.02	82.03	328.12	1,392.71

Joliff 1

Field name: Joliff 1

11/09/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 34 Plant date: 11/09/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/18/2023	640.00 *ton*	As-is		64.2	4,100.00	1,200.00	9,800.00		8.58

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	18.82	154.35	45.18	368.94	1,156.38

06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 34 Plant date: 06/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/03/2023	1,197.69 *ton*	As-is		68.7	3,000.00	900.00	4,100.00		6.19

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	35.23	211.36	63.41	288.85	1,364.99

Joliff 2

Field name: Joliff 2

11/09/2022: Oats, hay

Crop: Oats, hay Acres planted: 12 Plant date: 11/09/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/26/2023	54.00 *ton*	As-is		16.6	24,000.00	2,160.00	19,900.00		8.86

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	160.00	26.00	132.00	0.00
Total actual harvest content	4.50	216.00	19.44	179.10	665.03

06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 12 Plant date: 06/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/02/2023	398.25 *ton*	As-is		68.7	3,000.00	900.00	4,100.00		6.19

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	33.19	199.13	59.74	272.14	1,286.00

Mattos

Field name: Mattos

11/03/2022: Oats, hay

Crop: Oats, hay Acres planted: 18 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/26/2023	87.00 *ton*	As-is		8.0	17,000.00	2,750.00	25,900.00		9.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	160.00	26.00	132.00	0.00
Total actual harvest content	4.83	164.33	26.58	250.37	853.76

Mattos

06/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 18 Plant date: 06/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/02/2023	489.75 *ton*	As-is		67.2	4,000.00	1,000.00	3,400.00		4.67

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.21	217.67	54.42	185.02	833.53

N. Jackson

Field name: N. Jackson

11/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 19 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/18/2023	349.97 *ton*	As-is		72.6	5,600.00	1,300.00	10,500.00		11.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	18.42	206.30	47.89	386.81	1,170.89

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 19 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	675.68 *ton*	As-is		70.4	3,600.00	1,200.00	4,600.00		6.46

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	35.56	256.05	85.35	327.17	1,360.01

S. Jackson

Field name: S. Jackson

11/08/2022: Oats, hay

Crop: Oats, hay Acres planted: 9 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/26/2023	41.00 *ton*	As-is		16.6	24,000.00	2,160.00	19,900.00		8.86

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	160.00	26.00	132.00	0.00
Total actual harvest content	4.56	218.67	19.68	181.31	673.24

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 9 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/02/2023	256.92 *ton*	As-is		70.4	3,600.00	1,200.00	4,600.00		6.46

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.55	205.54	68.51	262.63	1,091.72

NUTRIENT BUDGET

A. LAND APPLICATIONS

Bacon - 11/02/2022: Oats, silage-soft dough

Field name: Bacon

Crop: Oats, silage-soft dough — Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/14/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Scraped Material	Corral solids	173.81	44.42	59.87	0.00	600.00 *ton*
Application event totals		173.81	44.42	59.87	0.00	

04/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 6	Surface water	0.00	0.00	0.00	60.84	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	60.84	

Bacon - 06/14/2023: Corn, silage

Field name: Bacon

Crop: Corn, silage — Plant date: 06/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/27/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	185.31	40.10	117.54	6,651.63	460.00 *ton*
Application event totals		185.31	40.10	117.54	6,651.63	

06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 6	Surface water	0.00	0.00	0.00	60.84	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	60.84	

Bacon - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 6	Surface water	0.00	0.00	0.00	55.77	4,455,000.00 *gal*
Application event totals		0.00	0.00	0.00	55.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	64.00	0.00	0.00	0.00	
Application event totals		64.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 6	Surface water	0.00	0.00	0.00	55.77	4,455,000.00 *gal*
Application event totals		0.00	0.00	0.00	55.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 6	Surface water	0.00	0.00	0.00	55.77	4,455,000.00 *gal*
Application event totals		0.00	0.00	0.00	55.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 6	Surface water	0.00	0.00	0.00	53.23	4,252,500.00 *gal*
Application event totals		0.00	0.00	0.00	53.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 6	Surface water	0.00	0.00	0.00	50.70	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	50.70	

Bacon - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 6	Surface water	0.00	0.00	0.00	50.70	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	50.70	

Boer - 11/02/2022: Oats, silage-soft dough

Field name: Boer

Crop: Oats, silage-soft dough

Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/14/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Scraped Material	Corral solids	175.10	44.75	60.31	0.00	170.00 *ton*
Application event totals		175.10	44.75	60.31	0.00	

04/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	65.34	1,215,000.00 *gal*
Application event totals		0.00	0.00	0.00	65.34	

Boer - 06/13/2023: Corn, silage

Field name: Boer

Crop: Corn, silage

Plant date: 06/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Boer - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/27/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	186.20	40.30	118.11	6,683.76	130.00 *ton*
Application event totals		186.20	40.30	118.11	6,683.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	87.12	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	87.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	65.34	1,215,000.00 *gal*
Application event totals		0.00	0.00	0.00	65.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	64.00	0.00	0.00	0.00	
Application event totals		64.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	65.34	1,215,000.00 *gal*
Application event totals		0.00	0.00	0.00	65.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	54.45	1,012,500.00 *gal*
Application event totals		0.00	0.00	0.00	54.45	

Boer - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	49.01	911,250.00 *gal*
Application event totals		0.00	0.00	0.00	49.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	43.56	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	43.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	43.56	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	43.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	43.56	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	43.56	

Connacher - 11/09/2022: Oats, hay

Field name: Connacher

Crop: Oats, hay

Plant date: 11/09/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	107.65	7.78	22.08	436.97	1,200,000.00 *gal*
MID Lateral 7	Surface water	3.49	0.00	0.00	61.45	8,100,000.00 *gal*
Application event totals		111.14	7.78	22.08	498.42	

Connacher - 11/09/2022: Oats, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	109.84	22.70	166.27	1,167.54	1,140,000.00 *gal*
MID Lateral 7	Surface water	0.00	0.00	0.00	84.65	7,695,000.00 *gal*
Application event totals		109.84	22.70	166.27	1,252.19	

Connacher - 06/15/2023: Corn, silage

Field name: Connacher

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	102.42	24.37	174.52	1,151.15	1,080,000.00 *gal*
MID Lateral 7	Surface water	0.00	0.00	0.00	80.19	7,290,000.00 *gal*
Application event totals		102.42	24.37	174.52	1,231.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	71.28	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	71.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	64.00	0.00	0.00	0.00	
Application event totals		64.00	0.00	0.00	0.00	

Connacher - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	71.28	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	71.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.74	11.42	62.82	559.87	900,000.00 *gal*
MID Lateral 7	Surface water	0.00	0.00	0.00	66.83	6,075,000.00 *gal*
Application event totals		52.74	11.42	62.82	626.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	69.05	6,277,500.00 *gal*
Application event totals		0.00	0.00	0.00	69.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.74	11.42	62.82	559.87	900,000.00 *gal*
MID Lateral 7	Surface water	0.00	0.00	0.00	66.83	6,075,000.00 *gal*
Application event totals		52.74	11.42	62.82	626.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	49.23	10.66	58.63	522.55	840,000.00 *gal*
MID Lateral 7	Surface water	0.00	0.00	0.00	62.37	5,670,000.00 *gal*
Application event totals		49.23	10.66	58.63	584.92	

Connacher - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	62.37	5,670,000.00 *gal*
Application event totals		0.00	0.00	0.00	62.37	

09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	62.37	5,670,000.00 *gal*
Application event totals		0.00	0.00	0.00	62.37	

Cover - 11/08/2022: Oats, silage-soft dough

Field name: Cover

Crop: Oats, silage-soft dough Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/14/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	118.42	8.56	24.28	480.67	570,000.00 *gal*
MID Lateral 7	Surface water	3.84	0.00	0.00	67.59	3,847,500.00 *gal*
Application event totals		122.25	8.56	24.28	548.27	

04/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	120.48	24.90	182.39	1,280.74	540,000.00 *gal*
MID Lateral 7	Surface water	0.00	0.00	0.00	92.85	3,645,000.00 *gal*
Application event totals		120.48	24.90	182.39	1,373.59	

Cover - 06/16/2023: Corn, silage

Cover - 06/16/2023: Corn, silage

Field name: Cover

Crop: Corn, silage

Plant date: 06/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	105.41	25.09	179.62	1,184.81	480,000.00 *gal*
MID Lateral 7	Surface water	0.00	0.00	0.00	82.54	3,240,000.00 *gal*
Application event totals		105.41	25.09	179.62	1,267.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	67.06	2,632,500.00 *gal*
Application event totals		0.00	0.00	0.00	67.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	64.00	0.00	0.00	0.00	
Application event totals		64.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	48.86	10.58	58.19	518.62	360,000.00 *gal*
MID Lateral 7	Surface water	0.00	0.00	0.00	61.90	2,430,000.00 *gal*
Application event totals		48.86	10.58	58.19	580.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.79	9.70	53.34	475.40	330,000.00 *gal*
MID Lateral 7	Surface water	0.00	0.00	0.00	56.74	2,227,500.00 *gal*
Application event totals		44.79	9.70	53.34	532.15	

Cover - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.75	9.26	50.91	453.79	315,000.00 *gal*
MID Lateral 7	Surface water	0.00	0.00	0.00	54.16	2,126,250.00 *gal*
Application event totals		42.75	9.26	50.91	507.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	51.59	2,025,000.00 *gal*
Application event totals		0.00	0.00	0.00	51.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	51.59	2,025,000.00 *gal*
Application event totals		0.00	0.00	0.00	51.59	

Joliff 1 - 11/09/2022: Oats, silage-soft dough

Field name: Joliff 1

Crop: Oats, silage-soft dough

Plant date: 11/09/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	97.52	7.05	20.00	395.85	840,000.00 *gal*
MID Lateral 7	Surface water	3.16	0.00	0.00	55.67	5,670,000.00 *gal*
Application event totals		100.68	7.05	20.00	451.51	

Joliff 1 - 11/09/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	104.73	21.65	158.55	1,113.32	840,000.00 *gal*
MID Lateral 7	Surface water	0.00	0.00	0.00	74.95	5,265,000.00 *gal*
Application event totals		104.73	21.65	158.55	1,188.27	

Joliff 1 - 06/14/2023: Corn, silage

Field name: Joliff 1

Crop: Corn, silage

Plant date: 06/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	88.36	21.03	150.56	993.15	720,000.00 *gal*
MID Lateral 7	Surface water	0.00	0.00	0.00	69.18	4,860,000.00 *gal*
Application event totals		88.36	21.03	150.56	1,062.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	63.42	4,455,000.00 *gal*
Application event totals		0.00	0.00	0.00	63.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	64.00	0.00	0.00	0.00	
Application event totals		64.00	0.00	0.00	0.00	

Joliff 1 - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	45.50	9.85	54.19	483.03	600,000.00 *gal*
MID Lateral 7	Surface water	0.00	0.00	0.00	57.65	4,050,000.00 *gal*
Application event totals		45.50	9.85	54.19	540.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	45.50	9.85	54.19	483.03	600,000.00 *gal*
MID Lateral 7	Surface water	0.00	0.00	0.00	57.65	4,050,000.00 *gal*
Application event totals		45.50	9.85	54.19	540.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	45.50	9.85	54.19	483.03	600,000.00 *gal*
MID Lateral 7	Surface water	0.00	0.00	0.00	57.65	4,050,000.00 *gal*
Application event totals		45.50	9.85	54.19	540.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	54.77	3,847,500.00 *gal*
Application event totals		0.00	0.00	0.00	54.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	51.89	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	51.89	

Joliff 2 - 11/09/2022: Oats, hay

Field name: Joliff 2

Crop: Oats, hay

Plant date: 11/09/2022

Joliff 2 - 11/09/2022: Oats, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	108.55	7.85	22.26	440.62	330,000.00 *gal*
MID Lateral 7	Surface water	3.52	0.00	0.00	61.96	2,227,500.00 *gal*
Application event totals		112.06	7.85	22.26	502.58	

04/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	105.98	21.91	160.43	1,126.58	300,000.00 *gal*
MID Lateral 7	Surface water	0.00	0.00	0.00	81.68	2,025,000.00 *gal*
Application event totals		105.98	21.91	160.43	1,208.25	

Joliff 2 - 06/14/2023: Corn, silage

Field name: Joliff 2

Crop: Corn, silage

Plant date: 06/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	83.45	19.86	142.20	937.98	240,000.00 *gal*
MID Lateral 7	Surface water	0.00	0.00	0.00	65.34	1,620,000.00 *gal*
Application event totals		83.45	19.86	142.20	1,003.32	

07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	57.17	1,417,500.00 *gal*
Application event totals		0.00	0.00	0.00	57.17	

Joliff 2 - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	64.00	0.00	0.00	0.00	
Application event totals		64.00	0.00	0.00	0.00	

07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	38.68	8.37	46.06	410.57	180,000.00 *gal*
MID Lateral 7	Surface water	0.00	0.00	0.00	49.01	1,215,000.00 *gal*
Application event totals		38.68	8.37	46.06	459.58	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	38.68	8.37	46.06	410.57	180,000.00 *gal*
MID Lateral 7	Surface water	0.00	0.00	0.00	49.01	1,215,000.00 *gal*
Application event totals		38.68	8.37	46.06	459.58	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	38.68	8.37	46.06	410.57	180,000.00 *gal*
MID Lateral 7	Surface water	0.00	0.00	0.00	49.01	1,215,000.00 *gal*
Application event totals		38.68	8.37	46.06	459.58	

08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	49.01	1,215,000.00 *gal*
Application event totals		0.00	0.00	0.00	49.01	

Joliff 2 - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	49.01	1,215,000.00 *gal*
Application event totals		0.00	0.00	0.00	49.01	

Mattos - 11/03/2022: Oats, hay

Field name: Mattos

Crop: Oats, hay

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/14/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Scraped Material	Corral solids	180.25	46.06	62.09	0.00	350.00 *ton*
Application event totals		180.25	46.06	62.09	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	70.79	2,632,500.00 *gal*
Application event totals		0.00	0.00	0.00	70.79	

Mattos - 06/13/2023: Corn, silage

Field name: Mattos

Crop: Corn, silage

Plant date: 06/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Mattos - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/27/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	186.20	40.30	118.11	6,683.76	260.00 *ton*
Application event totals		186.20	40.30	118.11	6,683.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	76.23	2,835,000.00 *gal*
Application event totals		0.00	0.00	0.00	76.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	65.34	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	65.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	64.00	0.00	0.00	0.00	
Application event totals		64.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	59.90	2,227,500.00 *gal*
Application event totals		0.00	0.00	0.00	59.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	59.90	2,227,500.00 *gal*
Application event totals		0.00	0.00	0.00	59.90	

Mattos - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	54.45	2,025,000.00 *gal*
Application event totals		0.00	0.00	0.00	54.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	54.45	2,025,000.00 *gal*
Application event totals		0.00	0.00	0.00	54.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	54.45	2,025,000.00 *gal*
Application event totals		0.00	0.00	0.00	54.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	54.45	2,025,000.00 *gal*
Application event totals		0.00	0.00	0.00	54.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	49.01	1,822,500.00 *gal*
Application event totals		0.00	0.00	0.00	49.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	49.01	1,822,500.00 *gal*
Application event totals		0.00	0.00	0.00	49.01	

Mattos - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	43.56	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	43.56	

N. Jackson - 11/08/2022: Oats, silage-soft dough

Field name: N. Jackson

Crop: Oats, silage-soft dough

Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/14/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	112.18	8.11	23.01	455.37	540,000.00 *gal*
MID Lateral 7	Surface water	3.63	0.00	0.00	64.04	3,645,000.00 *gal*
Application event totals		115.82	8.11	23.01	519.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	120.48	24.90	182.39	1,280.74	540,000.00 *gal*
MID Lateral 7	Surface water	0.00	0.00	0.00	92.85	3,645,000.00 *gal*
Application event totals		120.48	24.90	182.39	1,373.59	

N. Jackson - 06/15/2023: Corn, silage

Field name: N. Jackson

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

N. Jackson - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	98.82	23.52	168.39	1,110.76	450,000.00 *gal*
MID Lateral 7	Surface water	0.00	0.00	0.00	77.38	3,037,500.00 *gal*
Application event totals		98.82	23.52	168.39	1,188.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	72.22	2,835,000.00 *gal*
Application event totals		0.00	0.00	0.00	72.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	64.00	0.00	0.00	0.00	
Application event totals		64.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.93	11.46	63.04	561.84	390,000.00 *gal*
MID Lateral 7	Surface water	0.00	0.00	0.00	67.06	2,632,500.00 *gal*
Application event totals		52.93	11.46	63.04	628.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	48.86	10.58	58.19	518.62	360,000.00 *gal*
MID Lateral 7	Surface water	0.00	0.00	0.00	61.90	2,430,000.00 *gal*
Application event totals		48.86	10.58	58.19	580.52	

N. Jackson - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	48.86	10.58	58.19	518.62	360,000.00 *gal*
MID Lateral 7	Surface water	0.00	0.00	0.00	61.90	2,430,000.00 *gal*
Application event totals		48.86	10.58	58.19	580.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	56.74	2,227,500.00 *gal*
Application event totals		0.00	0.00	0.00	56.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	61.90	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	61.90	

S. Jackson - 11/08/2022: Oats, hay

Field name: S. Jackson

Crop: Oats, hay

Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/14/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	111.84	8.09	22.93	453.97	255,000.00 *gal*
MID Lateral 7	Surface water	3.62	0.00	0.00	63.84	1,721,250.00 *gal*
Application event totals		115.46	8.09	22.93	517.81	

S. Jackson - 11/08/2022: Oats, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	113.05	23.37	171.13	1,201.68	240,000.00 *gal*
MID Lateral 7	Surface water	0.00	0.00	0.00	87.12	1,620,000.00 *gal*
Application event totals		113.05	23.37	171.13	1,288.80	

S. Jackson - 06/15/2023: Corn, silage

Field name: S. Jackson

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	97.36	23.17	165.90	1,094.31	210,000.00 *gal*
MID Lateral 7	Surface water	0.00	0.00	0.00	76.23	1,417,500.00 *gal*
Application event totals		97.36	23.17	165.90	1,170.54	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	65.34	1,215,000.00 *gal*
Application event totals		0.00	0.00	0.00	65.34	

07/18/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	64.00	0.00	0.00	0.00	
Application event totals		64.00	0.00	0.00	0.00	

S. Jackson - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.98	9.30	51.18	456.19	150,000.00 *gal*
MID Lateral 7	Surface water	0.00	0.00	0.00	54.45	1,012,500.00 *gal*
Application event totals		42.98	9.30	51.18	510.64	

07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	38.68	8.37	46.06	410.57	135,000.00 *gal*
MID Lateral 7	Surface water	0.00	0.00	0.00	49.01	911,250.00 *gal*
Application event totals		38.68	8.37	46.06	459.58	

08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.38	7.44	40.95	364.95	120,000.00 *gal*
MID Lateral 7	Surface water	0.00	0.00	0.00	43.56	810,000.00 *gal*
Application event totals		34.38	7.44	40.95	408.52	

08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	54.45	1,012,500.00 *gal*
Application event totals		0.00	0.00	0.00	54.45	

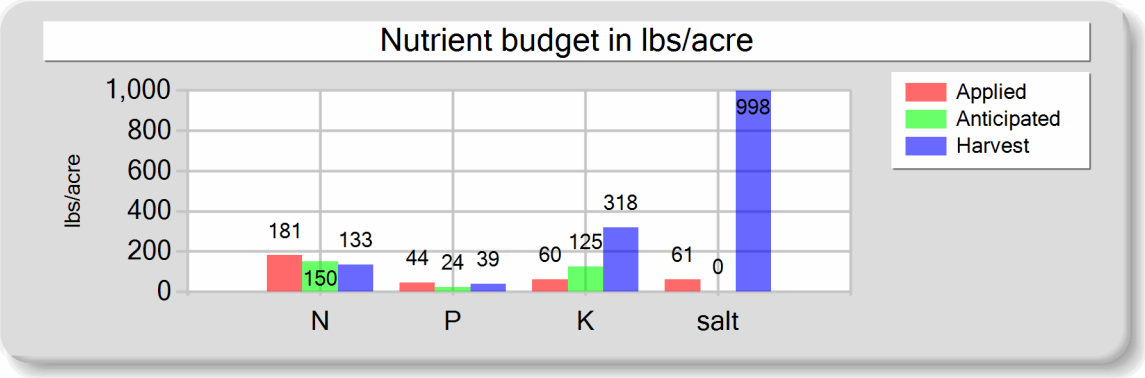
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 7	Surface water	0.00	0.00	0.00	54.45	1,012,500.00 *gal*
Application event totals		0.00	0.00	0.00	54.45	

B. NUTRIENT BUDGET

Bacon - 11/02/2022: Oats, silage-soft dough

Field name: Bacon | Crop: Oats, silage-soft dough | Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	173.81	44.42	59.87	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	60.84
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	180.81	44.42	59.87	60.84
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	133.23	38.99	318.46	998.15
Nutrient balance	47.58	5.42	-258.59	-937.31
Applied to removed ratio	1.36	1.14	0.19	0.06

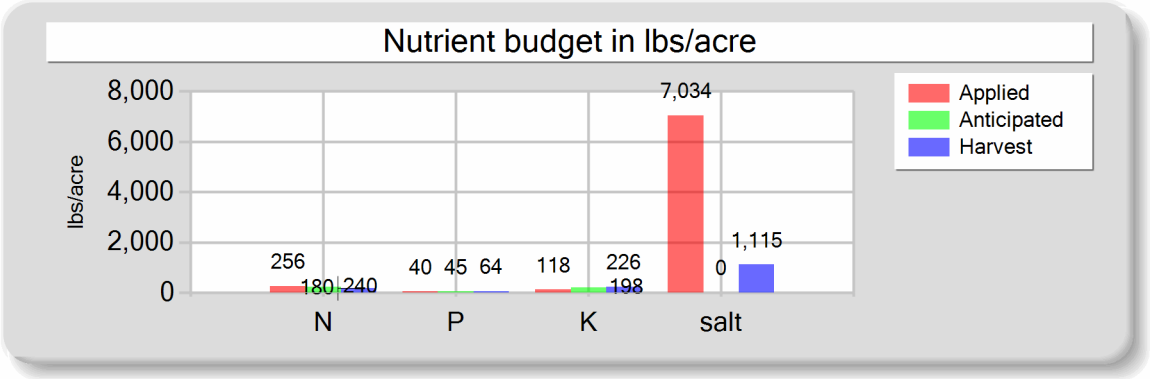
Fresh water applied	
4,860,000.00	*gallons*
178.98	*acre-inches*
5.59	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Bacon - 06/14/2023: Corn, silage

Field name: Bacon Crop: Corn, silage Plant date: 06/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	185.31	40.10	117.54	6,651.63
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	382.75
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	256.31	40.10	117.54	7,034.38
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	179.87	63.83	226.29	1,114.61
Nutrient balance	76.43	-23.72	-108.75	5,919.77
Applied to removed ratio	1.42	0.63	0.52	6.31

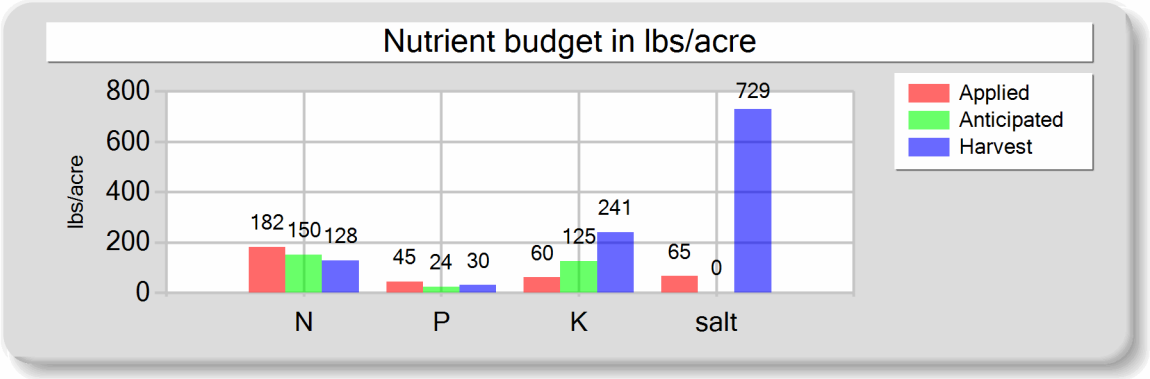
Fresh water applied	
30,577,500.00	*gallons*
1,126.07	*acre-inches*
35.19	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Boer - 11/02/2022: Oats, silage-soft dough

Field name: Boer | Crop: Oats, silage-soft dough | Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	175.10	44.75	60.31	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	65.34
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	182.10	44.75	60.31	65.34
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	128.38	29.80	240.71	728.63
Nutrient balance	53.72	14.95	-180.39	-663.29
Applied to removed ratio	1.42	1.50	0.25	0.09

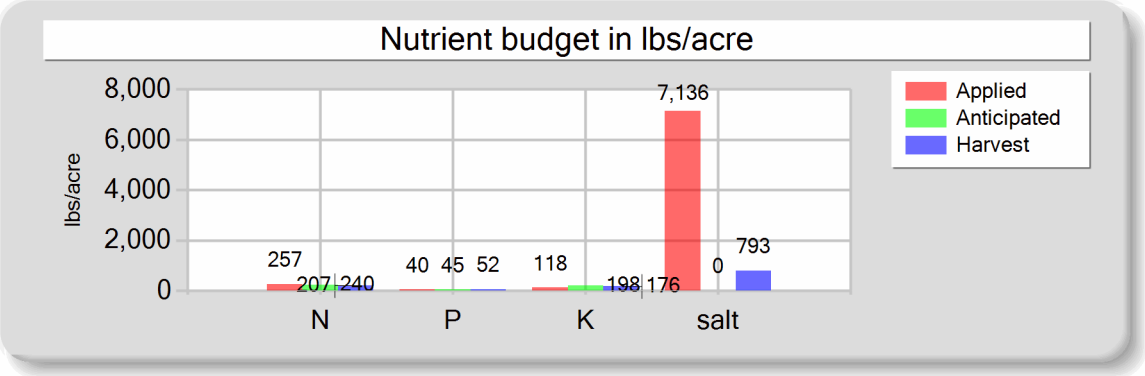
Fresh water applied	
1,215,000.00	*gallons*
44.74	*acre-inches*
4.97	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Boer - 06/13/2023: Corn, silage

Field name: Boer | Crop: Corn, silage | Plant date: 06/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	186.20	40.30	118.11	6,683.76
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	451.94
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	257.20	40.30	118.11	7,135.71
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	207.00	51.75	175.95	792.67
Nutrient balance	50.20	-11.45	-57.83	6,343.04
Applied to removed ratio	1.24	0.78	0.67	9.00

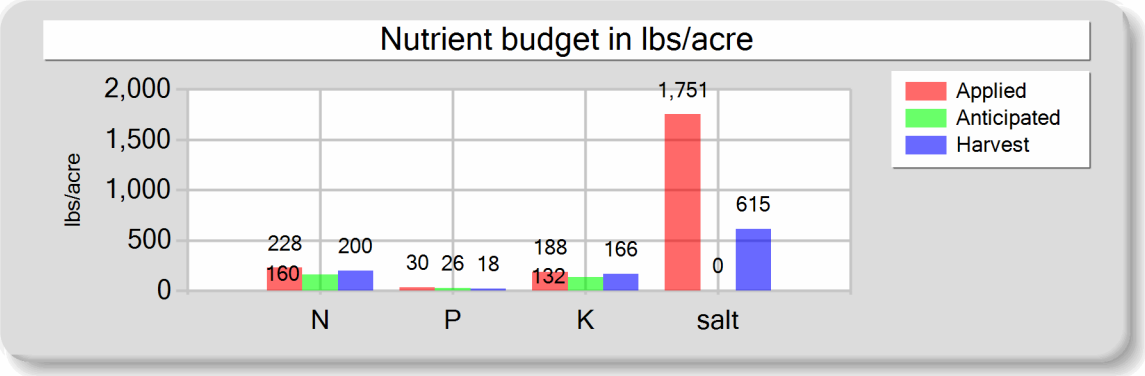
Fresh water applied	
8,403,750.00	*gallons*
309.48	*acre-inches*
34.39	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Connacher - 11/09/2022: Oats, hay

Field name: Connacher Crop: Oats, hay Plant date: 11/09/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	217.49	30.49	188.34	1,604.52
Fresh water	3.49	0.00	0.00	146.10
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	227.97	30.49	188.34	1,750.61
Anticipated crop nutrient removal	160.00	26.00	132.00	0.00
Actual crop nutrient removal	199.64	17.97	165.53	614.65
Nutrient balance	28.34	12.52	22.81	1,135.96
Applied to removed ratio	1.14	1.70	1.14	2.85

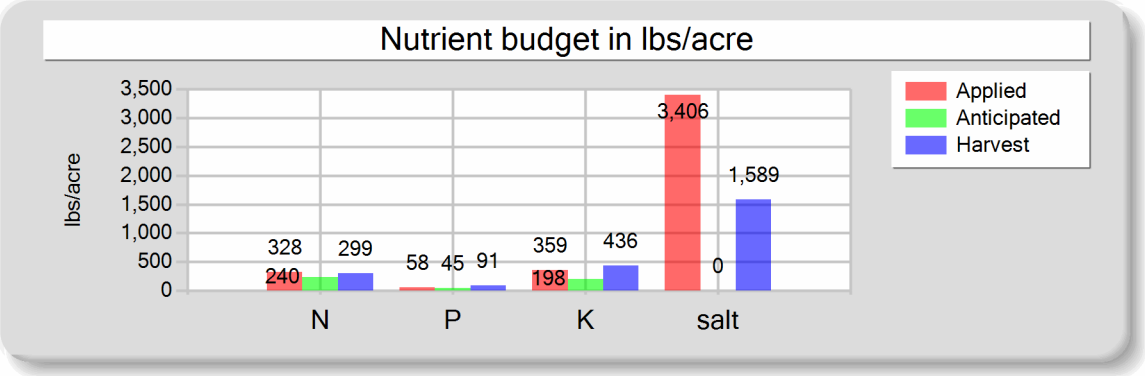
Fresh water applied	
15,795,000.00	*gallons*
581.68	*acre-inches*
13.22	*inches/acre*

Process wastewater applied	
2,340,000.00	*gallons*
86.17	*acre-inches*
1.96	*inches/acre*

Total harvests for the crop	
1	*harvests*

Connacher - 06/15/2023: Corn, silage

Field name: Connacher Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	257.13	57.87	358.77	2,793.45
Fresh water	0.00	0.00	0.00	612.58
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	328.13	57.87	358.77	3,406.03
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	299.28	91.09	435.91	1,589.43
Nutrient balance	28.85	-33.21	-77.14	1,816.59
Applied to removed ratio	1.10	0.64	0.82	2.14

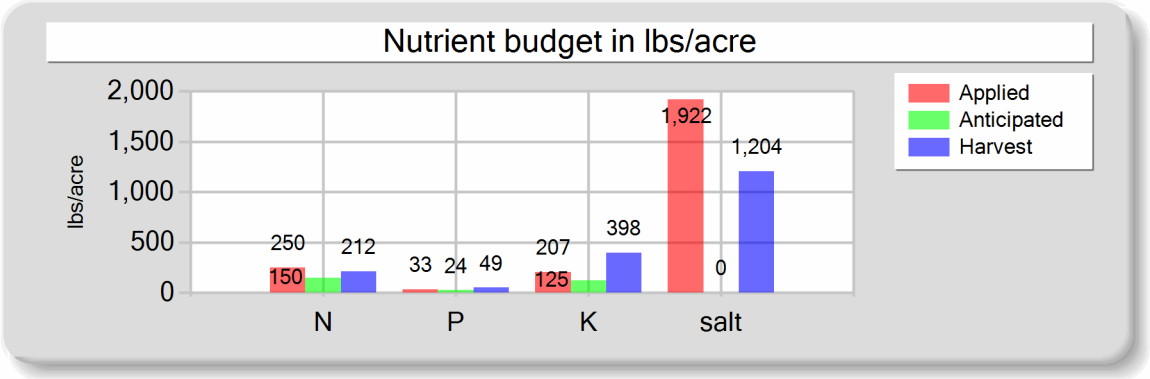
Fresh water applied	
55,687,500.00	*gallons*
2,050.78	*acre-inches*
46.61	*inches/acre*

Process wastewater applied	
3,720,000.00	*gallons*
136.99	*acre-inches*
3.11	*inches/acre*

Total harvests for the crop	
1	*harvests*

Cover - 11/08/2022: Oats, silage-soft dough

Field name: Cover | Crop: Oats, silage-soft dough | Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	238.90	33.47	206.67	1,761.41
Fresh water	3.84	0.00	0.00	160.45
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	249.74	33.47	206.67	1,921.86
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	212.19	49.26	397.86	1,204.35
Nutrient balance	37.54	-15.79	-191.19	717.51
Applied to removed ratio	1.18	0.68	0.52	1.60

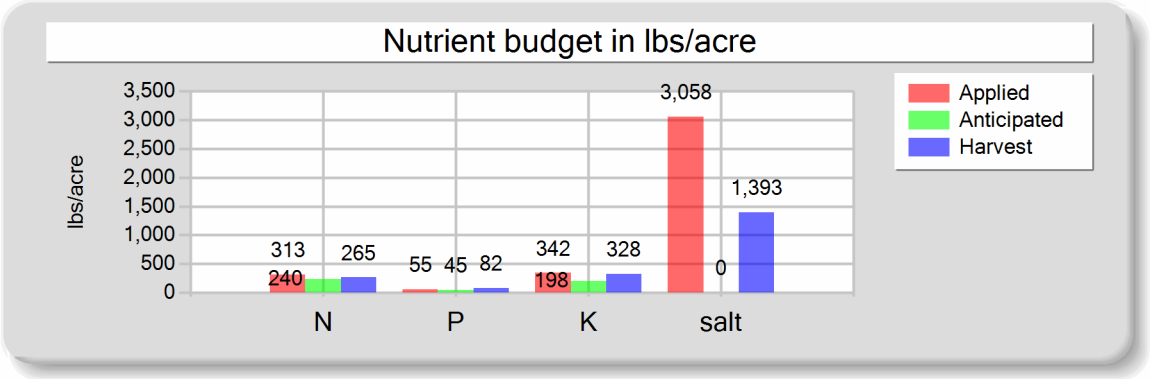
Fresh water applied	
7,492,500.00	*gallons*
275.92	*acre-inches*
14.52	*inches/acre*

Process wastewater applied	
1,110,000.00	*gallons*
40.88	*acre-inches*
2.15	*inches/acre*

Total harvests for the crop	
1	*harvests*

Cover - 06/16/2023: Corn, silage

Field name: Cover | Crop: Corn, silage | Plant date: 06/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	241.81	54.62	342.06	2,632.63
Fresh water	0.00	0.00	0.00	425.58
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	312.81	54.62	342.06	3,058.21
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	265.02	82.03	328.12	1,392.71
Nutrient balance	47.79	-27.41	13.94	1,665.50
Applied to removed ratio	1.18	0.67	1.04	2.20

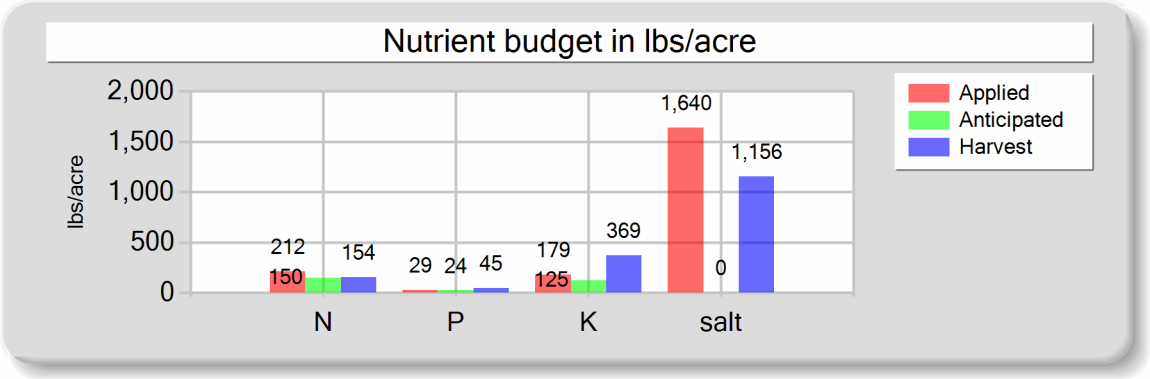
Fresh water applied	
16,706,250.00	*gallons*
615.23	*acre-inches*
32.38	*inches/acre*

Process wastewater applied	
1,485,000.00	*gallons*
54.69	*acre-inches*
2.88	*inches/acre*

Total harvests for the crop	
1	*harvests*

Joliff 1 - 11/09/2022: Oats, silage-soft dough

Field name: Joliff 1 Crop: Oats, silage-soft dough Plant date: 11/09/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	202.25	28.70	178.54	1,509.17
Fresh water	3.16	0.00	0.00	130.62
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	212.41	28.70	178.54	1,639.79
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	154.35	45.18	368.94	1,156.38
Nutrient balance	58.06	-16.48	-190.40	483.40
Applied to removed ratio	1.38	0.64	0.48	1.42

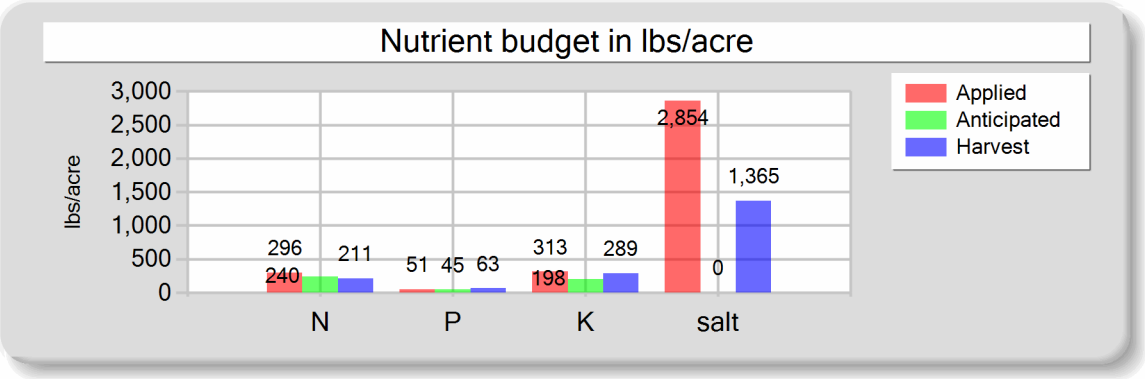
Fresh water applied	
10,935,000.00	*gallons*
402.70	*acre-inches*
11.84	*inches/acre*

Process wastewater applied	
1,680,000.00	*gallons*
61.87	*acre-inches*
1.82	*inches/acre*

Total harvests for the crop	
1	*harvests*

Joliff 1 - 06/14/2023: Corn, silage

Field name: Joliff 1 Crop: Corn, silage Plant date: 06/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	224.87	50.59	313.14	2,442.24
Fresh water	0.00	0.00	0.00	412.23
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	295.87	50.59	313.14	2,854.46
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	211.36	63.41	288.85	1,364.99
Nutrient balance	84.52	-12.82	24.29	1,489.47
Applied to removed ratio	1.40	0.80	1.08	2.09

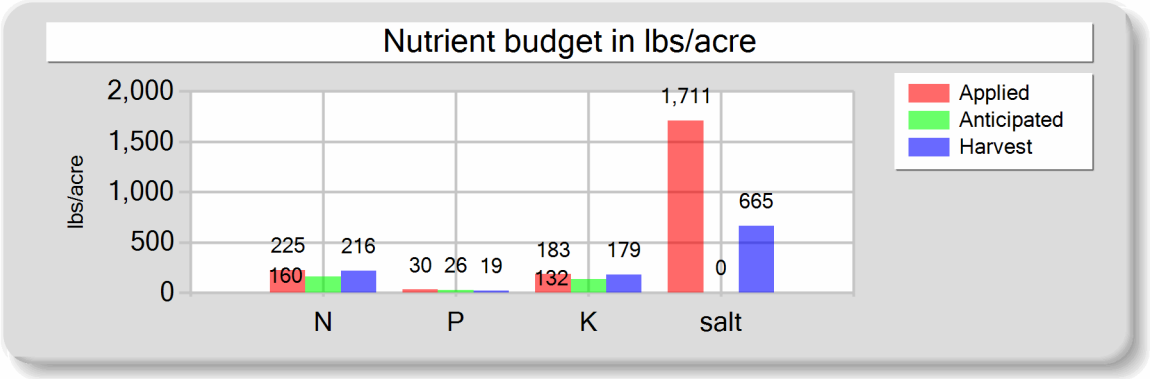
Fresh water applied	
28,957,500.00	*gallons*
1,066.41	*acre-inches*
31.36	*inches/acre*

Process wastewater applied	
2,520,000.00	*gallons*
92.80	*acre-inches*
2.73	*inches/acre*

Total harvests for the crop	
1	*harvests*

Joliff 2 - 11/09/2022: Oats, hay

Field name: Joliff 2 Crop: Oats, hay Plant date: 11/09/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	214.53	29.75	182.69	1,567.19
Fresh water	3.52	0.00	0.00	143.64
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	225.05	29.75	182.69	1,710.83
Anticipated crop nutrient removal	160.00	26.00	132.00	0.00
Actual crop nutrient removal	216.00	19.44	179.10	665.03
Nutrient balance	9.05	10.31	3.59	1,045.80
Applied to removed ratio	1.04	1.53	1.02	2.57

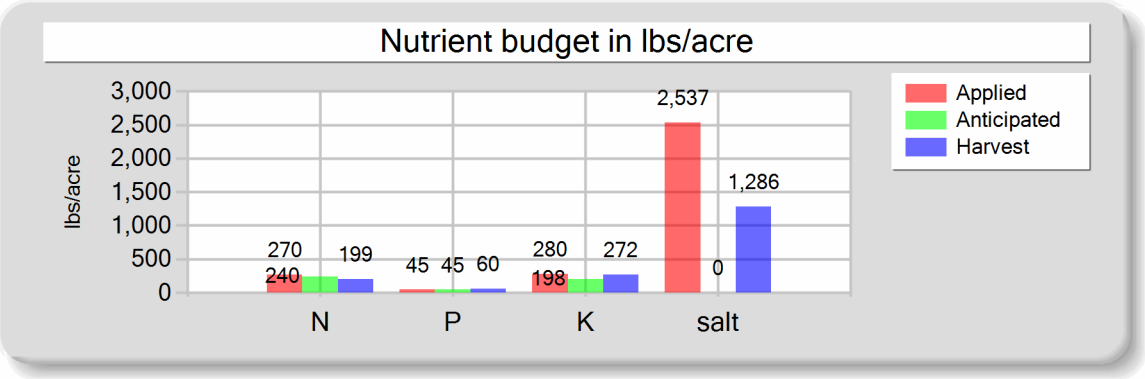
Fresh water applied	
4,252,500.00	*gallons*
156.61	*acre-inches*
13.05	*inches/acre*

Process wastewater applied	
630,000.00	*gallons*
23.20	*acre-inches*
1.93	*inches/acre*

Total harvests for the crop	
1	*harvests*

Joliff 2 - 06/14/2023: Corn, silage

Field name: Joliff 2 Crop: Corn, silage Plant date: 06/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	199.49	44.98	280.39	2,169.70
Fresh water	0.00	0.00	0.00	367.55
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	270.49	44.98	280.39	2,537.25
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	199.13	59.74	272.14	1,286.00
Nutrient balance	71.36	-14.75	8.25	1,251.25
Applied to removed ratio	1.36	0.75	1.03	1.97

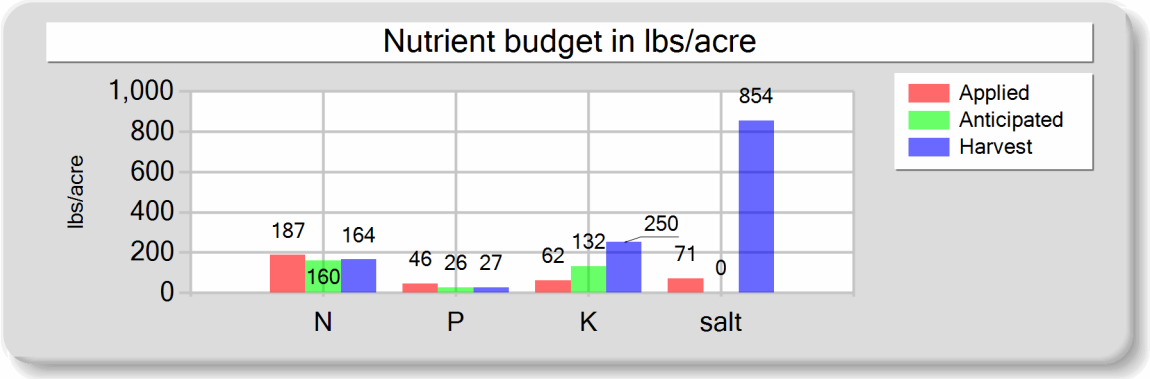
Fresh water applied	
9,112,500.00	*gallons*
335.58	*acre-inches*
27.97	*inches/acre*

Process wastewater applied	
780,000.00	*gallons*
28.72	*acre-inches*
2.39	*inches/acre*

Total harvests for the crop	
1	*harvests*

Mattos - 11/03/2022: Oats, hay

Field name: Mattos Crop: Oats, hay Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	180.25	46.06	62.09	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	70.79
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	187.25	46.06	62.09	70.79
Anticipated crop nutrient removal	160.00	26.00	132.00	0.00
Actual crop nutrient removal	164.33	26.58	250.37	853.76
Nutrient balance	22.92	19.48	-188.28	-782.97
Applied to removed ratio	1.14	1.73	0.25	0.08

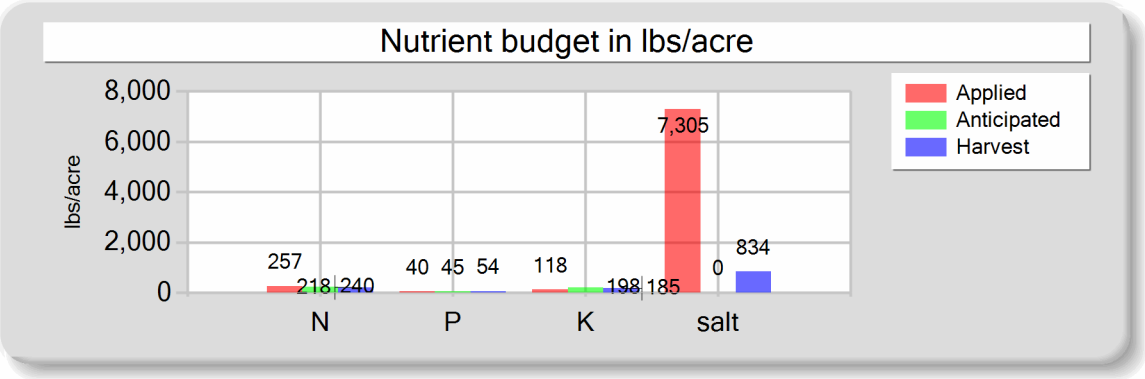
Fresh water applied	
2,632,500.00	*gallons*
96.95	*acre-inches*
5.39	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Mattos - 06/13/2023: Corn, silage

Field name: Mattos | Crop: Corn, silage | Plant date: 06/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	186.20	40.30	118.11	6,683.76
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	620.74
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	257.20	40.30	118.11	7,304.51
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	217.67	54.42	185.02	833.53
Nutrient balance	39.53	-14.12	-66.90	6,470.97
Applied to removed ratio	1.18	0.74	0.64	8.76

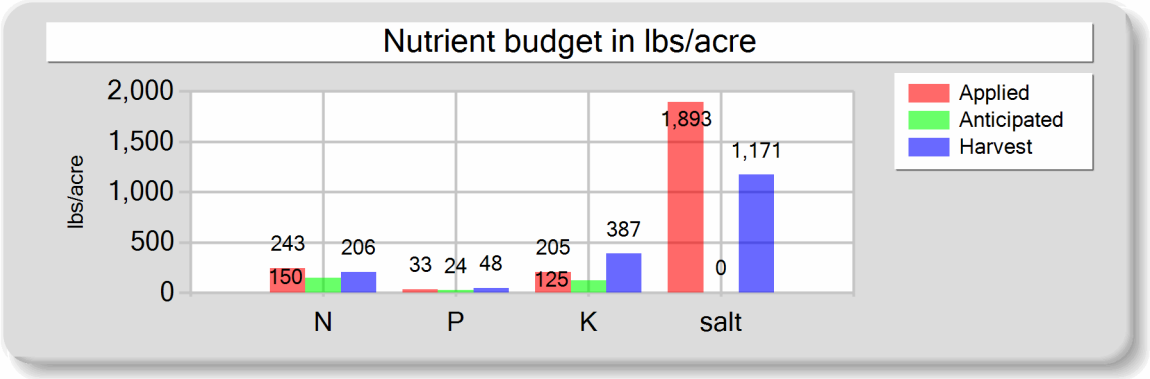
Fresh water applied	
23,085,000.00	*gallons*
850.14	*acre-inches*
47.23	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

N. Jackson - 11/08/2022: Oats, silage-soft dough

Field name: N. Jackson Crop: Oats, silage-soft dough Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	232.67	33.01	205.39	1,736.11
Fresh water	3.63	0.00	0.00	156.89
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	243.30	33.01	205.39	1,893.00
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	206.30	47.89	386.81	1,170.89
Nutrient balance	37.00	-14.88	-181.42	722.11
Applied to removed ratio	1.18	0.69	0.53	1.62

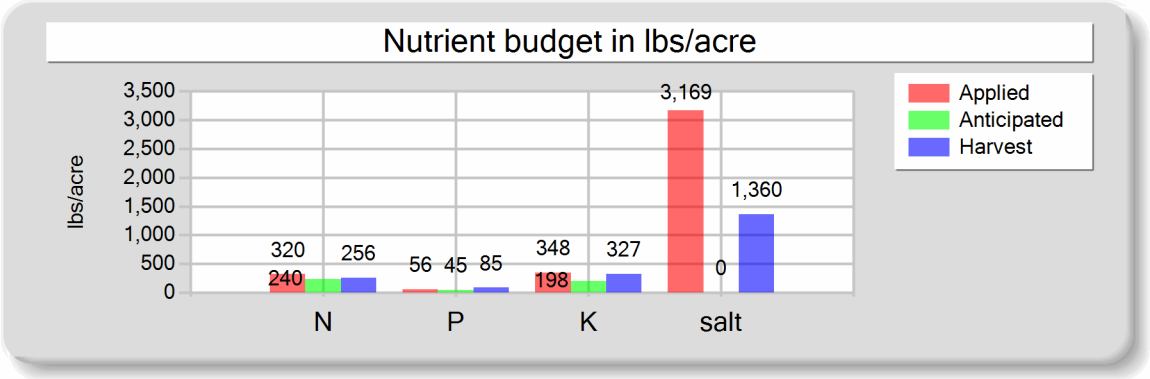
Fresh water applied	
7,290,000.00	*gallons*
268.47	*acre-inches*
14.13	*inches/acre*

Process wastewater applied	
1,080,000.00	*gallons*
39.77	*acre-inches*
2.09	*inches/acre*

Total harvests for the crop	
1	*harvests*

N. Jackson - 06/15/2023: Corn, silage

Field name: N. Jackson Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	249.47	56.14	347.80	2,709.84
Fresh water	0.00	0.00	0.00	459.11
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	320.47	56.14	347.80	3,168.95
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	256.05	85.35	327.17	1,360.01
Nutrient balance	64.42	-29.21	20.63	1,808.94
Applied to removed ratio	1.25	0.66	1.06	2.33

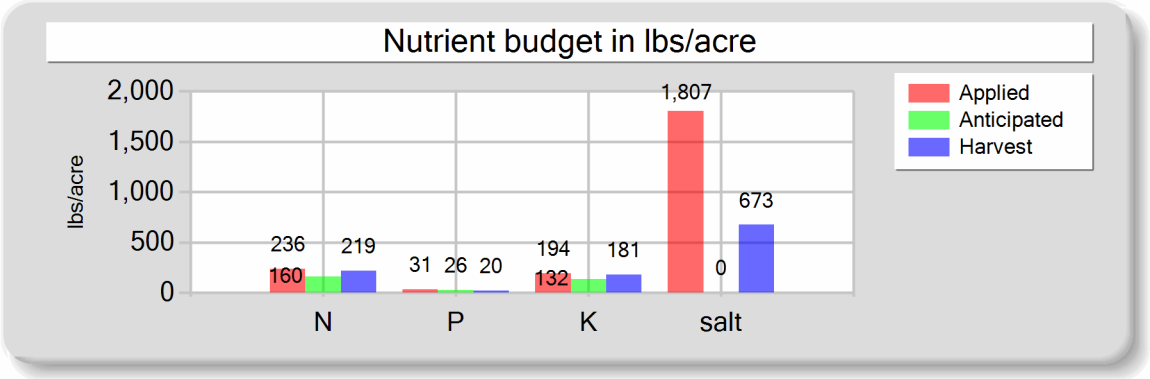
Fresh water applied	
18,022,500.00	*gallons*
663.71	*acre-inches*
34.93	*inches/acre*

Process wastewater applied	
1,560,000.00	*gallons*
57.45	*acre-inches*
3.02	*inches/acre*

Total harvests for the crop	
1	*harvests*

S. Jackson - 11/08/2022: Oats, hay

Field name: S. Jackson Crop: Oats, hay Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	224.88	31.45	194.06	1,655.65
Fresh water	3.62	0.00	0.00	150.96
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	235.51	31.45	194.06	1,806.61
Anticipated crop nutrient removal	160.00	26.00	132.00	0.00
Actual crop nutrient removal	218.67	19.68	181.31	673.24
Nutrient balance	16.84	11.77	12.75	1,133.37
Applied to removed ratio	1.08	1.60	1.07	2.68

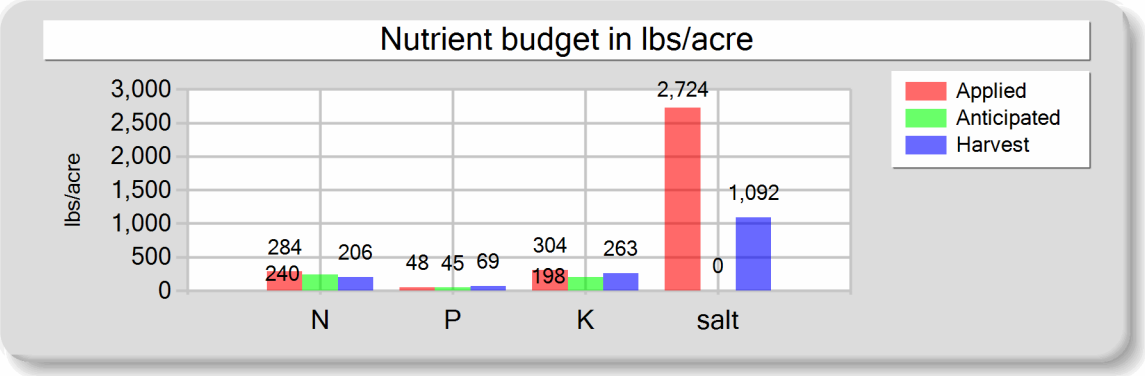
Fresh water applied	
3,341,250.00	*gallons*
123.05	*acre-inches*
13.67	*inches/acre*

Process wastewater applied	
495,000.00	*gallons*
18.23	*acre-inches*
2.03	*inches/acre*

Total harvests for the crop	
1	*harvests*

S. Jackson - 06/15/2023: Corn, silage

Field name: S. Jackson Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	64.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	213.40	48.29	304.09	2,326.03
Fresh water	0.00	0.00	0.00	397.49
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	284.40	48.29	304.09	2,723.52
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	205.54	68.51	262.63	1,091.72
Nutrient balance	78.86	-20.22	41.46	1,631.81
Applied to removed ratio	1.38	0.70	1.16	2.49

Fresh water applied	
7,391,250.00	*gallons*
272.19	*acre-inches*
30.24	*inches/acre*

Process wastewater applied	
615,000.00	*gallons*
22.65	*acre-inches*
2.52	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Stacked

Sample and source description: Stacked

Sample date: 10/04/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 48.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,000.00	2,300.00	3,100.00	15,300.00	3,700.00	600.00	1,900.00			0.00
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00			0.01

Spring Manure

Sample and source description: Spring Manure

Sample date: 06/02/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 51.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,400.00	2,900.00	8,500.00							48.10
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

WWQ4

Sample and source description: WWQ4

Sample date: 11/13/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	473.00	106.00	0.00	0.00	34.20	97.00								2,610.00	1,920
DL	0.01	0.01	0.50	0.50	0.01	0.01								0.01	1

WWQ1

Sample and source description: WWQ1

Sample date: 04/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.28

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	508.00	352.00	0.00	0.00	105.00	769.00								8,140.00	5,400
DL	10.00	2.00	0.50	0.50	0.20	0.50								0.10	10

WWQ2

Sample and source description: WWQ2

Sample date: 06/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.46

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	500.00	382.00	0.00	0.00	119.00	852.00								8,470.00	5,620
DL	10.00	2.00	0.50	0.50	0.20	0.50								1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 09/25/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	309.00	224.00	0.00	0.00	66.90	368.00								4,940.00	3,280
DL	10.00	2.00	0.50	0.50	0.20	0.50								0.10	10

WWQ4

Sample and source description: WWQ4

Sample date: 10/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.20

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	441.00	371.00	0.00	0.00	77.70	656.00								6,310.00	4,190
DL	10.00	2.00	0.50	0.50	0.20	0.50								0.10	10

C. FRESH WATER ANALYSES

MID Lateral 6

MID Lateral 6

Sample description: MID Lateral 6

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								38.80	48
DL	1.00	0.50	0.50								10.00	10

MID Lateral 7

MID lateral 7

Sample description: MID lateral 7

Sample date: 07/22/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.27	0.00	0.00								165.00	40
DL	0.50	0.50	0.50								1.00	10

MID Lateral 7

Sample description: MID Lateral 7

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								43.50	58
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Bacon - 11/02/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,100.00	1,200.00	9,800.00		8.58
DL	0.05	0.02	0.02		0.05

Bacon - 06/14/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,100.00	1,100.00	3,900.00		5.52
DL	0.05	0.02	0.02		0.05

Boer - 11/02/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,600.00	1,300.00	10,500.00		11.60
DL	0.05	0.02	0.02		0.05

Boer - 06/13/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,000.00	1,000.00	3,400.00		4.67
DL	0.05	0.02	0.02		0.05

Connacher - 11/09/2022: Oats, hay

Forage

Sample and source description: Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 16.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,000.00	2,160.00	19,900.00		8.86
DL	100.00	100.00	100.00		0.01

Connacher - 06/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,600.00	1,400.00	6,700.00		7.02
DL	0.05	0.02	0.02		0.05

Cover - 11/08/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,600.00	1,300.00	10,500.00		11.60
DL	0.05	0.02	0.02		0.05

Cover - 06/16/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,200.00	1,300.00	5,200.00		6.53
DL	0.05	0.02	0.02		0.05

Joliff 1 - 11/09/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,100.00	1,200.00	9,800.00		8.58
DL	0.05	0.02	0.02		0.05

Joliff 1 - 06/14/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,000.00	900.00	4,100.00		6.19
DL	0.05	0.02	0.02		0.05

Joliff 2 - 11/09/2022: Oats, hay

Oat Silage

Sample and source description: Oat Silage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 16.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,000.00	2,160.00	19,900.00		8.86
DL	100.00	100.00	100.00		0.01

Joliff 2 - 06/14/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,000.00	900.00	4,100.00		6.19
DL	0.05	0.02	0.02		0.05

Mattos - 11/03/2022: Oats, hay

Oats

Sample and source description: Oats

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 8.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,000.00	2,750.00	25,900.00		9.60
DL	100.00	100.00	100.00		0.01

Mattos - 06/13/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,000.00	1,000.00	3,400.00		4.67
DL	0.05	0.02	0.02		0.05

N. Jackson - 11/08/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,600.00	1,300.00	10,500.00		11.60
DL	0.05	0.02	0.02		0.05

N. Jackson - 06/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,600.00	1,200.00	4,600.00		6.46
DL	0.05	0.02	0.02		0.05

S. Jackson - 11/08/2022: Oats, hay

Oat Hay

Sample and source description: Oat Hay

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 16.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,000.00	2,160.00	19,900.00		8.86
DL	100.00	100.00	100.00		0.01

S. Jackson - 06/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,600.00	1,200.00	4,600.00		6.46
DL	0.05	0.02	0.02		0.05

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

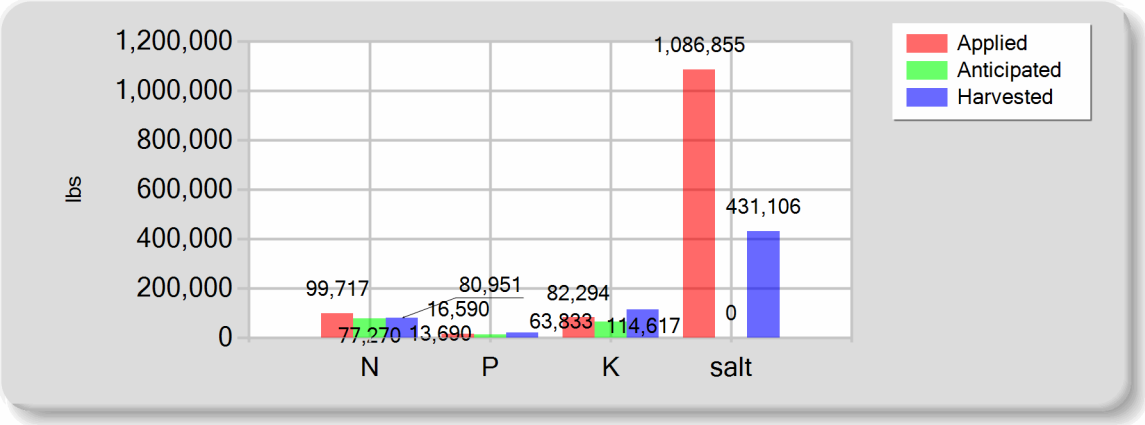
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

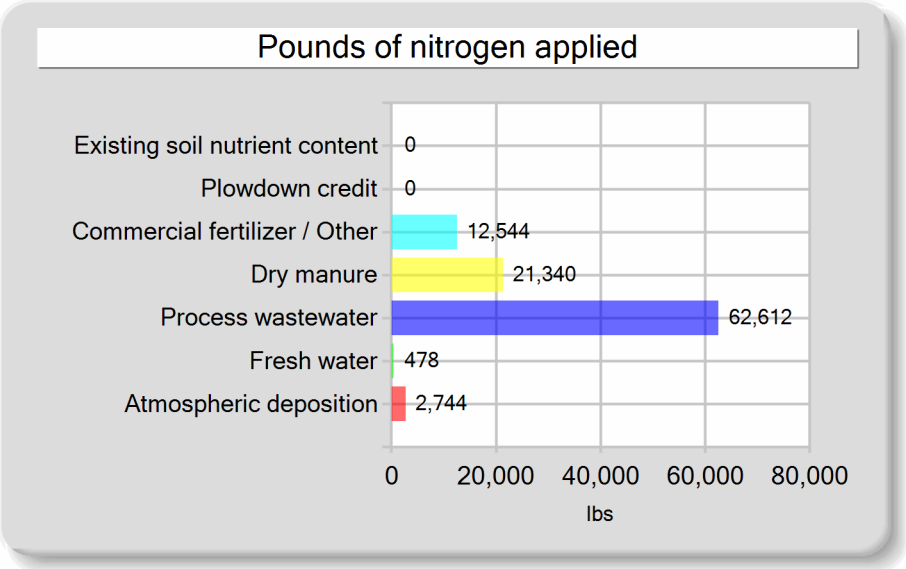
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

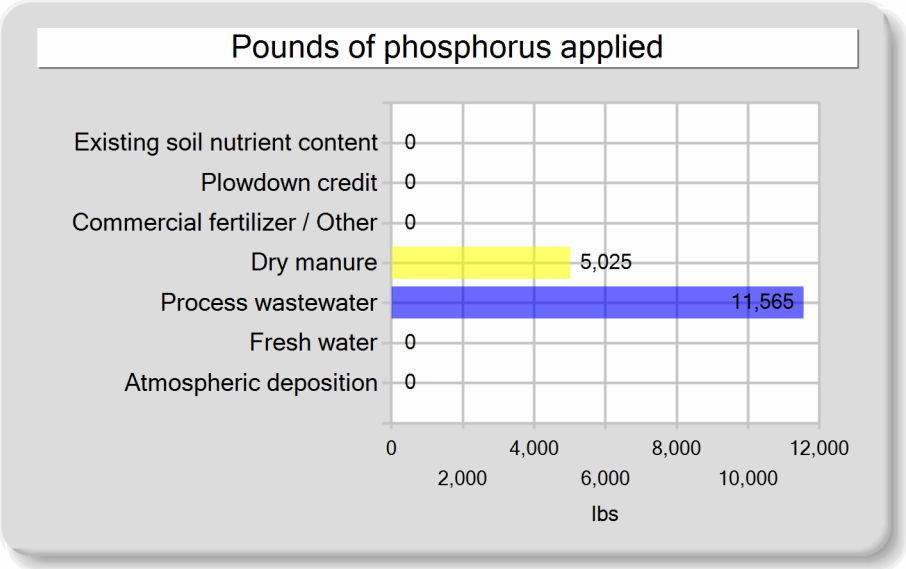
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	12,544.00	0.00	0.00	0.00
Dry manure	21,339.58	5,024.61	10,526.61	393,313.70
Process wastewater	62,612.16	11,565.39	71,767.46	576,495.97
Fresh water	477.58	0.00	0.00	117,044.95
Atmospheric deposition	2,744.00	0.00	0.00	0.00
Total nutrients applied	99,717.32	16,590.00	82,294.07	1,086,854.61
Anticipated crop nutrient removal	77,270.00	13,690.00	63,832.50	0.00
Actual crop nutrient removal	80,951.31	20,742.34	114,617.04	431,105.80
Nutrient balance	18,766.01	-4,152.33	-32,322.97	655,748.81
Applied to removed ratio	1.23	0.80	0.72	2.52

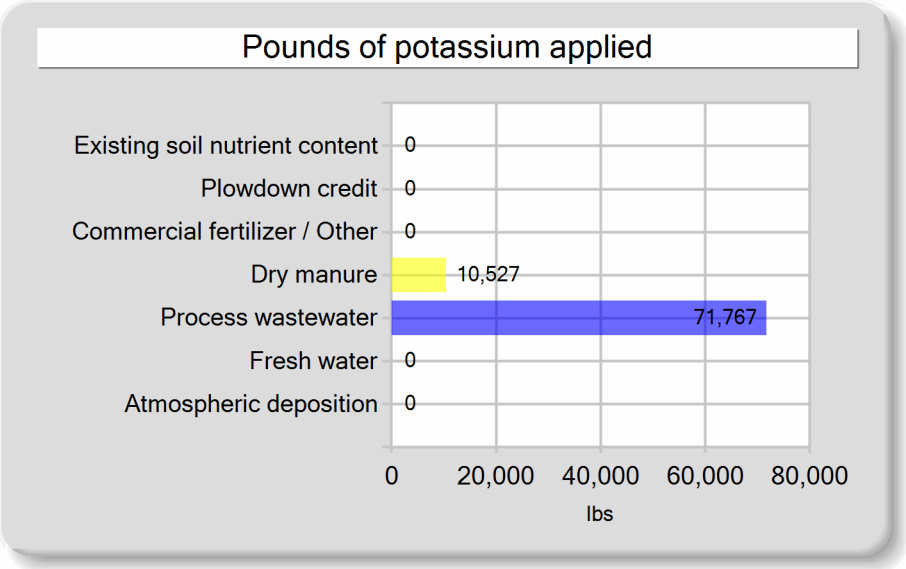
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

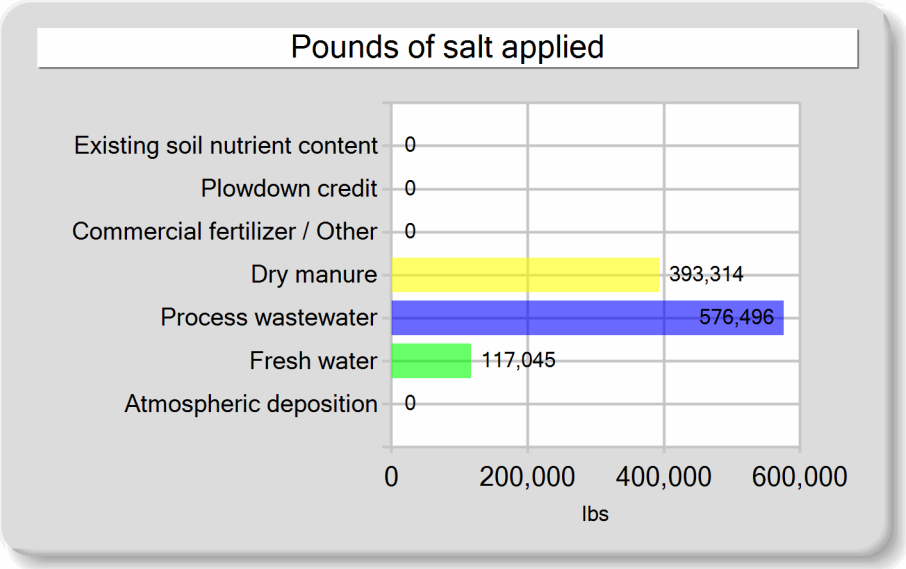


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Due to an oversight, Domestic Well samples were not procured during the 2023 reporting period.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Ivan Wyeth II (Jun 10, 2024 20:31 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Ivan Wyeth, Jr	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 10, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

2023 Annual Report Wyeth Dairy

Final Audit Report 2024-06-11

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